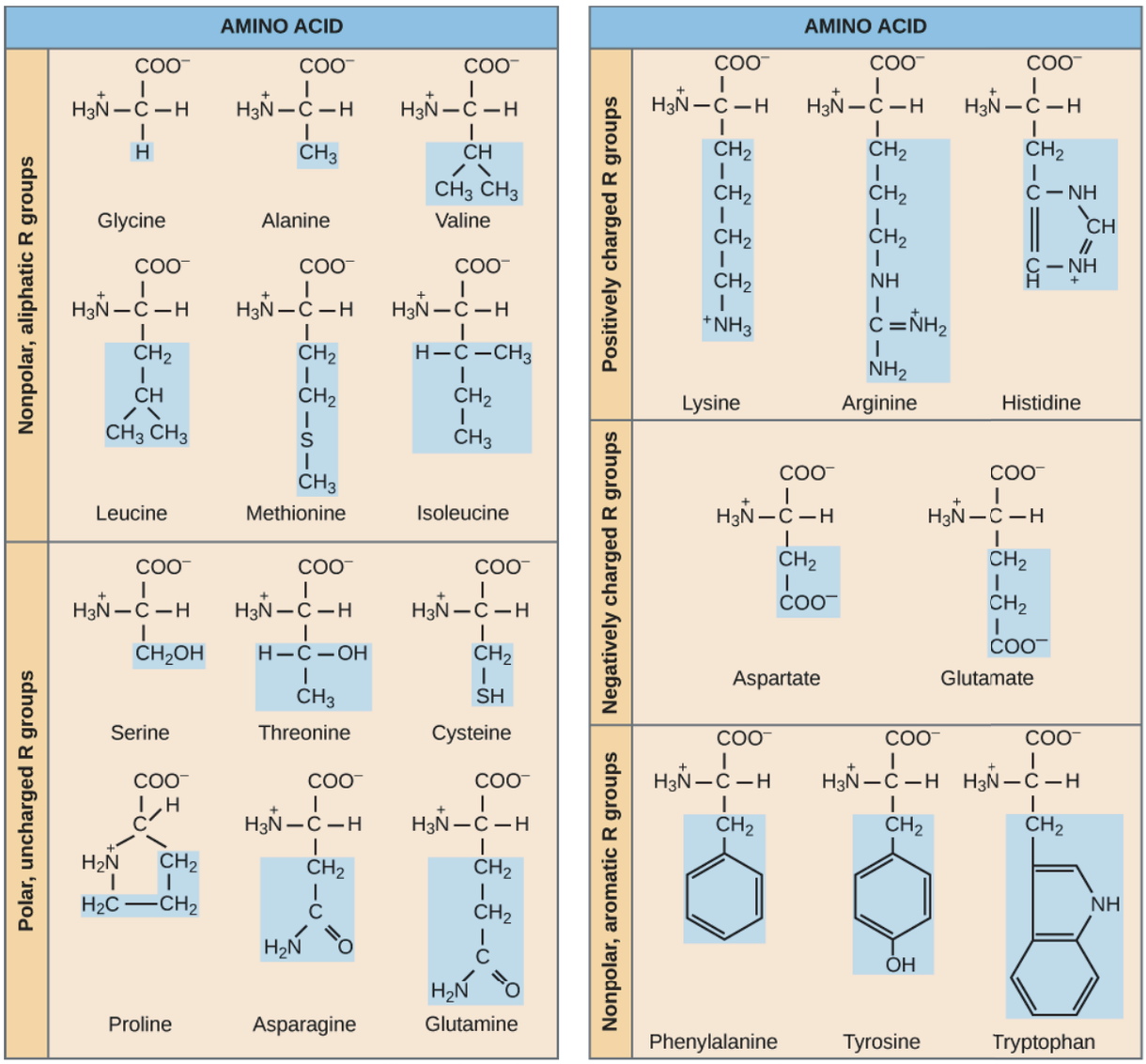


Figure 8.11(b): Classification of Amino Acids

8.5.2 Structure of Protein



Protein are synthesised on the ribosome as a linear sequence of amino acids which are held together by peptide bonds. After synthesis, the protein attains conformational change into a specific 3D form for proper functioning. According to the mode of folding, four levels of protein organisation have been recognised namely primary, secondary, tertiary and quaternary (Figure 8.12).

- The **primary structure** is linear arrangement of amino acids in a polypeptide chain.
- **Secondary structure** arises when various functional groups are exposed on outer surface of the molecular interaction by forming hydrogen bonds. This causes the aminoacid chain to twist into coiled configuration called α-**helix** or to fold into a flat β-pleated sheets.
- **Tertiary protein structure** arises when the secondary level proteins fold into globular structure called domains.
- **Quaternary protein structure** may be assumed by some complex proteins in which more than one polypeptide forms a large multiunit protein. The individual polypeptide chains of the protein are called **subunits** and the active protein itself is called a **multimer.**

XIth_Botany_Chapter_8_EM.indd 184 05-01-2023 18:27:36